U. S . Crime Data Analysis

Exploring Patterns and insights for Crime prevention

SITE LINK: https://sites.google.com/view/crime-trends-in-the-usa/home?authuser=1

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Problem statement

Why is crime analysis important?

- Understanding crime trends can help in preventive measures and policy recommendations.

Key questions to address:

- What are the most common types of crimes, and how do they vary across regions?
- Are crime rates increasing or decreasing over time?
- How do socio-economic factors influence crime trends?

Objective:

- Using historical crime data to uncover trends and patterns that can inform decision-making.

Data Overview

- **Dataset:** U. S . Crime Data
- **Source:** Kaggle
- **Number of records:** 638454
- **Time period covered:** 1980 – 2014
- **Key features in the dataset:**
  - Crime types: Murder, robbery, assault, burglary, etc.
  - Geographic information: Crime occurrences in different states/regions.
  - Time related features: Yearly data distribution.
- This comprehensive data allows us to examine trends, regional variations and predictive factors.

Data Profiling:

Column Name	Data Type	Missing Values	Unique Values	Example Values
Record ID	Integer	0	638,454	1, 2, 3, 4, 5
Agency Code	Text	0	12,003	AK00101, AK00103, AK00106
Agency Name	Text	0	9,216	Anchorage, Juneau, Nome
Agency Type	Text	0	7	Municipal Police, County Police
City	Text	0	1,782	Anchorage, Nome, Bethel
State	Text	0	51	Alaska, California, Texas
Year	Integer	0	35	1980, 1981, 1982
Month	Text	0	12	January, March, June
Incident ID	Integer	0	1,000	1, 2, 3, 4, 41
Crime Type	Text	0	2	Murder, Manslaughter by Negligence
Crime Solved	Text	0	2	Yes, No
Victim Sex	Text	0	3	Male, Female, Unknown
Victim Age	Integer	0	101	14, 30, 42, 99
Victim Race	Text	0	5	White, Black, Asian, Native American
Victim Ethnicity	Text	0	3	Hispanic, Non-Hispanic, Unknown
Perpetrator Sex	Text	0	3	Male, Female, Unknown
Perpetrator Age	Mixed (Text)	0	191	15, 27, 42, 0
Perpetrator Race	Text	0	5	White, Black, Asian, Native American
Perpetrator Ethnicity	Text	0	3	Hispanic, Non-Hispanic, Unknown
Relationship	Text	0	28	Stranger, Wife, Girlfriend, Acquaintance
Weapon	Text	0	16	Handgun, Knife, Blunt Object, Poison
Victim Count	Integer	0	11	0, 1, 2, 3, 6
Perpetrator Count	Integer	0	11	0, 1, 2, 3, 4

Data processing and cleaning:

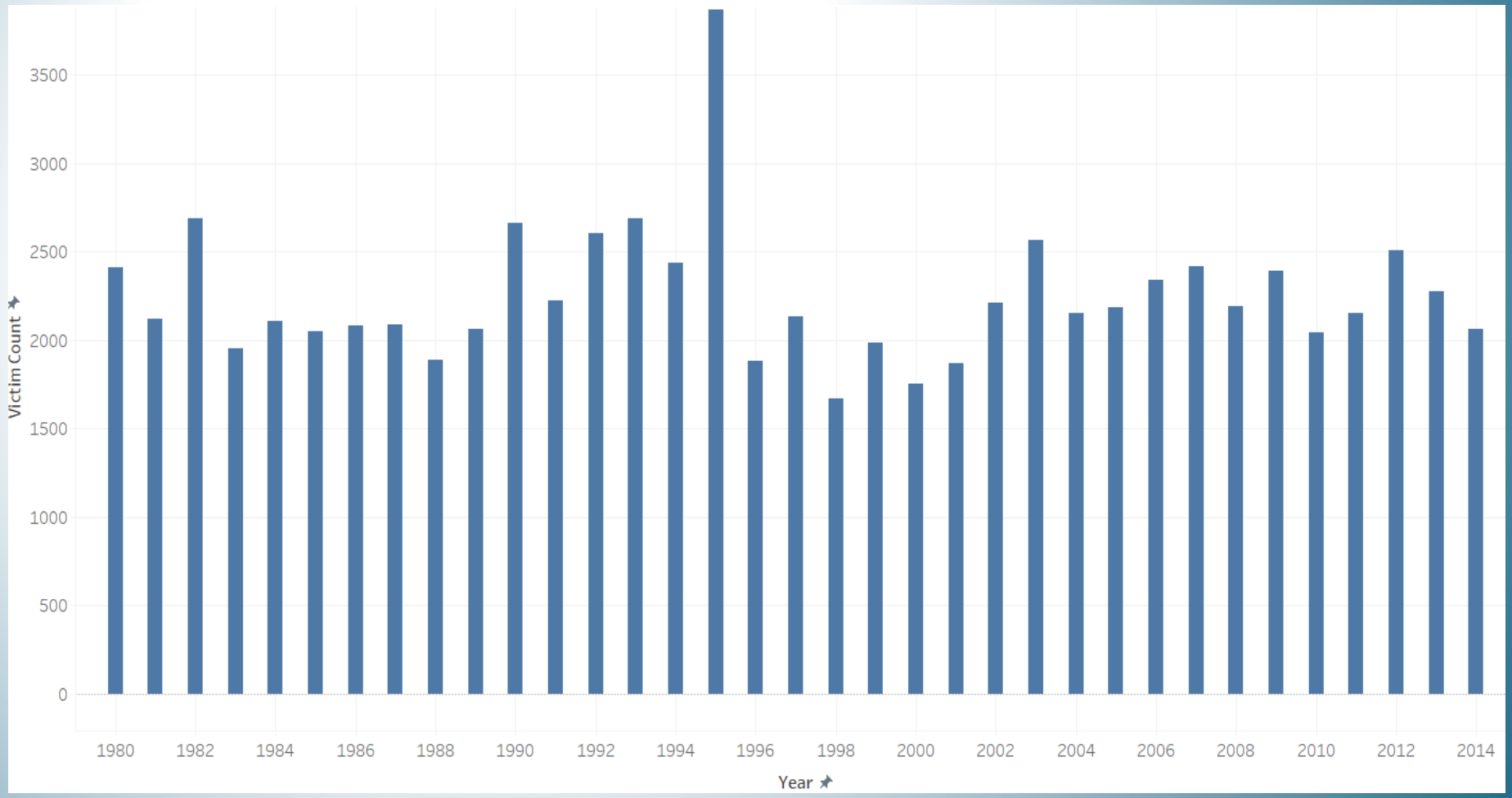
- Key variables like Crime Type and Weapon were converted using one-hot encoding for analysis.
- To ensure consistency, we fixed data types—converting ‘Year’ to integer and properly handling ‘Victim Count’.
- Finally, the cleaned dataset was saved in an organized folder, ready for visualization and insights.

Exploratory Data Analysis (EDA)

- **Descriptive Statistics:**
  - Mean, median, and standard deviation of crime types across states.
  - Most frequently occurring crimes.
- **Distribution Analysis:**
  - Histograms & Box Plots: Visualizing the spread of crime rates across states.
  - Correlation Analysis (Heatmap): Understanding relationships between crime types.
- **Initial Observations:**
  - Certain states consistently report higher crime rates.
  - Some crimes have strong seasonal trends.
  - Possible correlation between crime rates and population density.

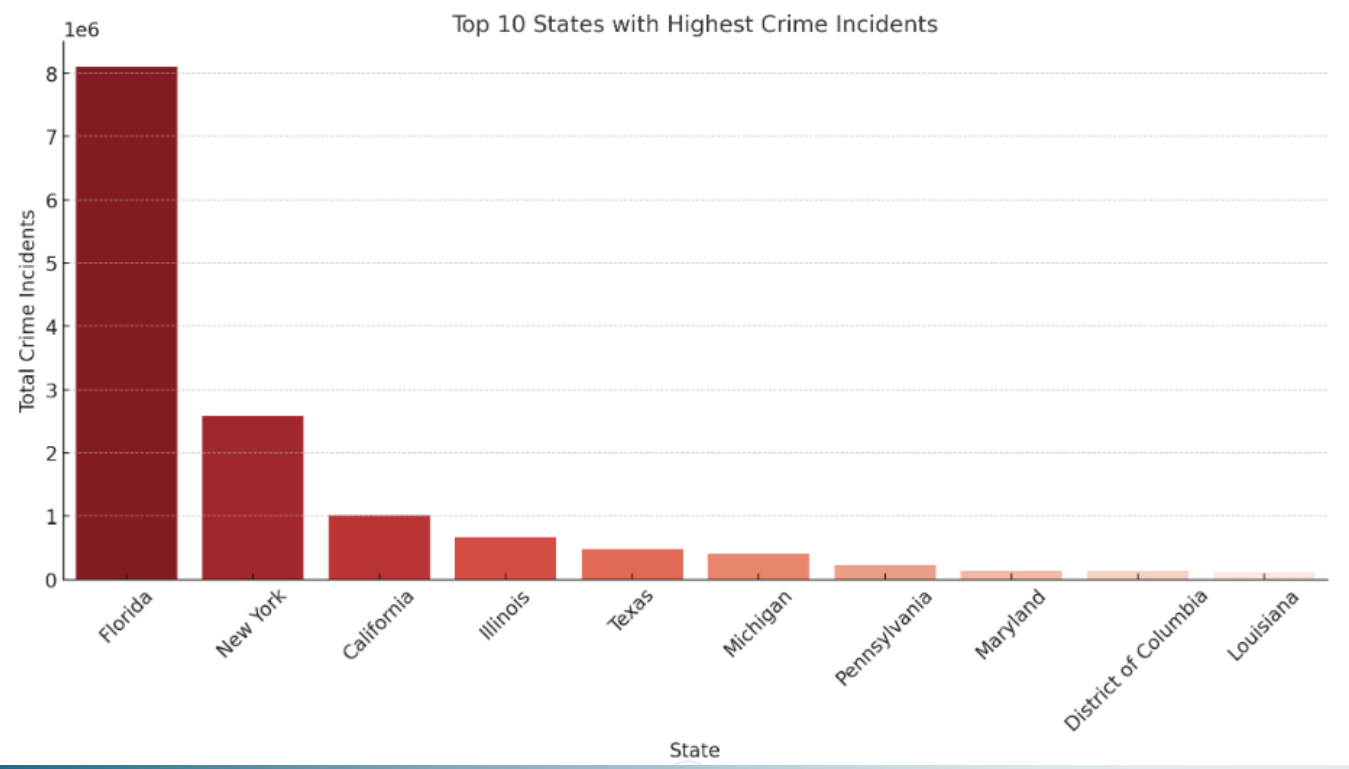
Key crime trends and patterns

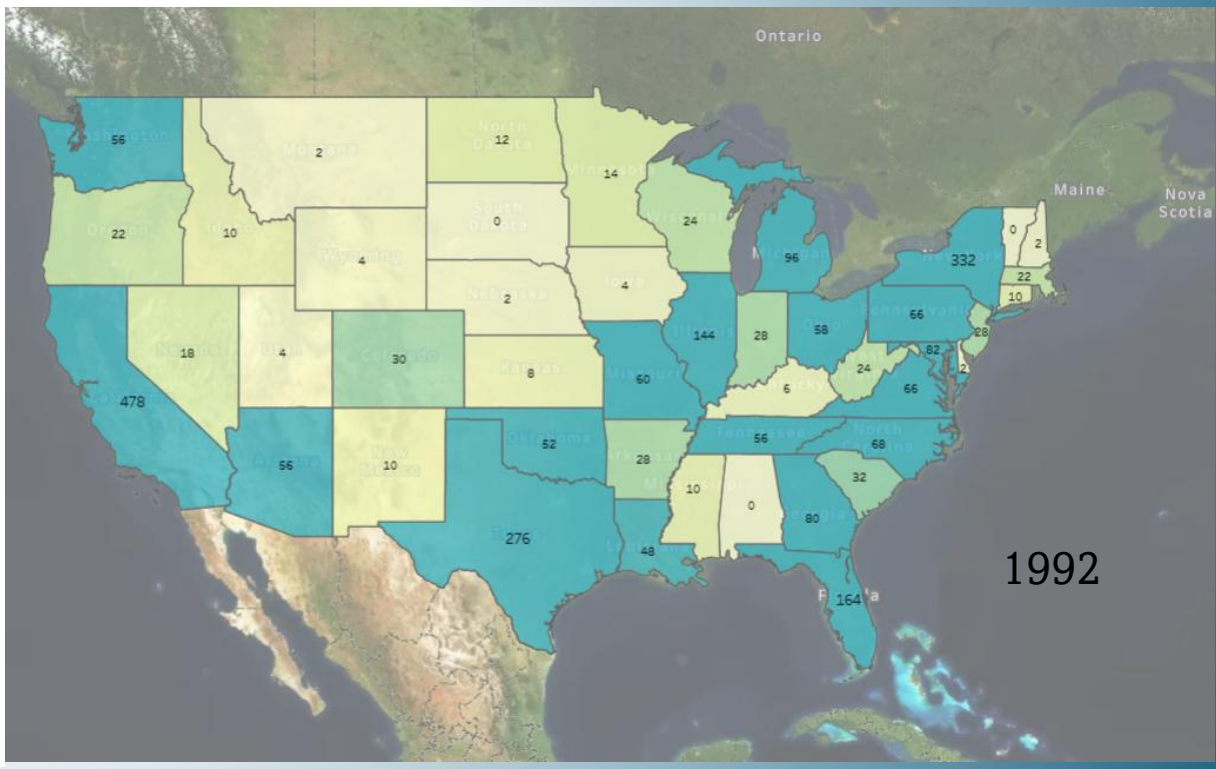
Overall Crime Trends over the years: Crime incidents peaked in certain years and declined in others, indicating fluctuations in crime rates.



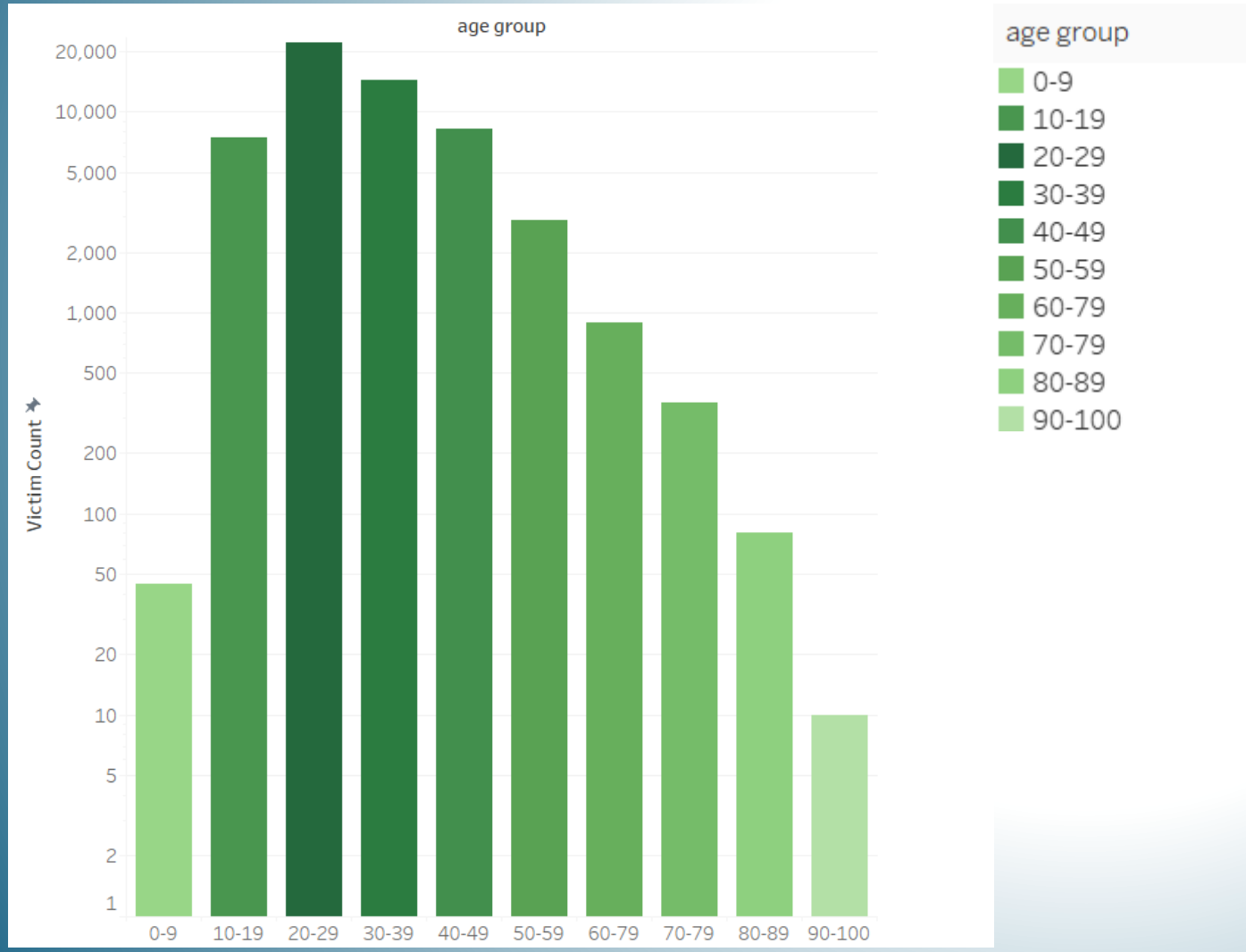
The bar chart below highlights the top 10 states with the highest crime incidents while the heatmap shows crime rates according to states in the year 1992:

- Certain states report significantly higher crime rates than others.
- The variations may be due to population density, law enforcement efficiency, or socio-economic conditions.





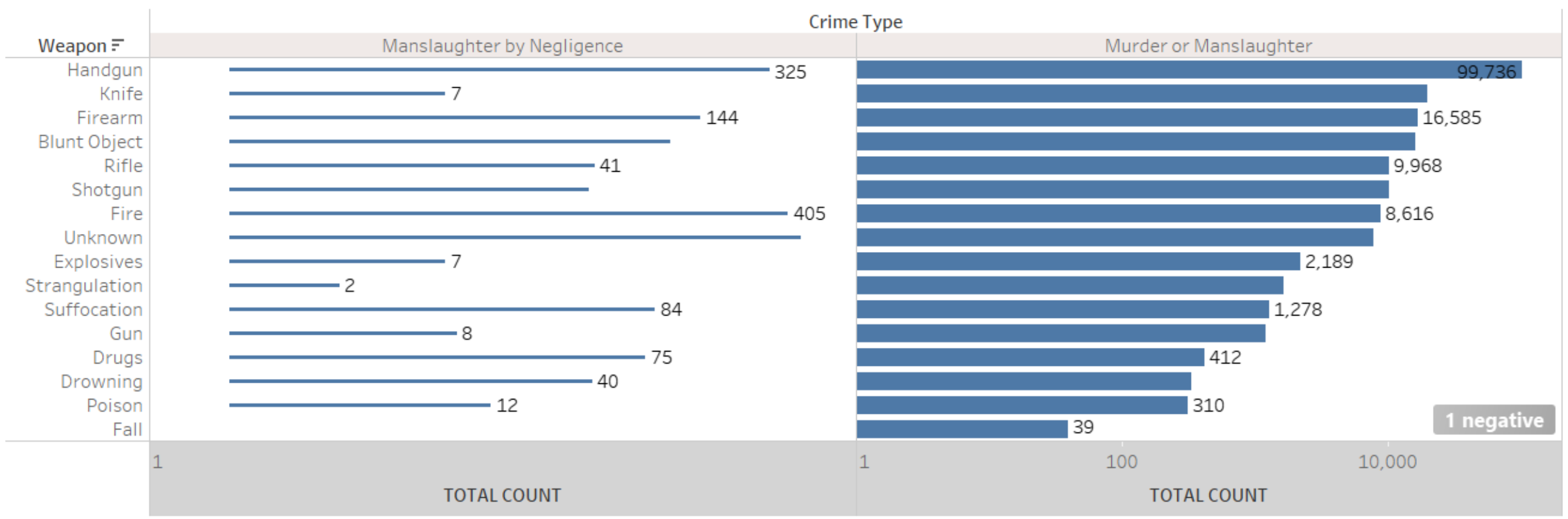
Victim distribution by age group:



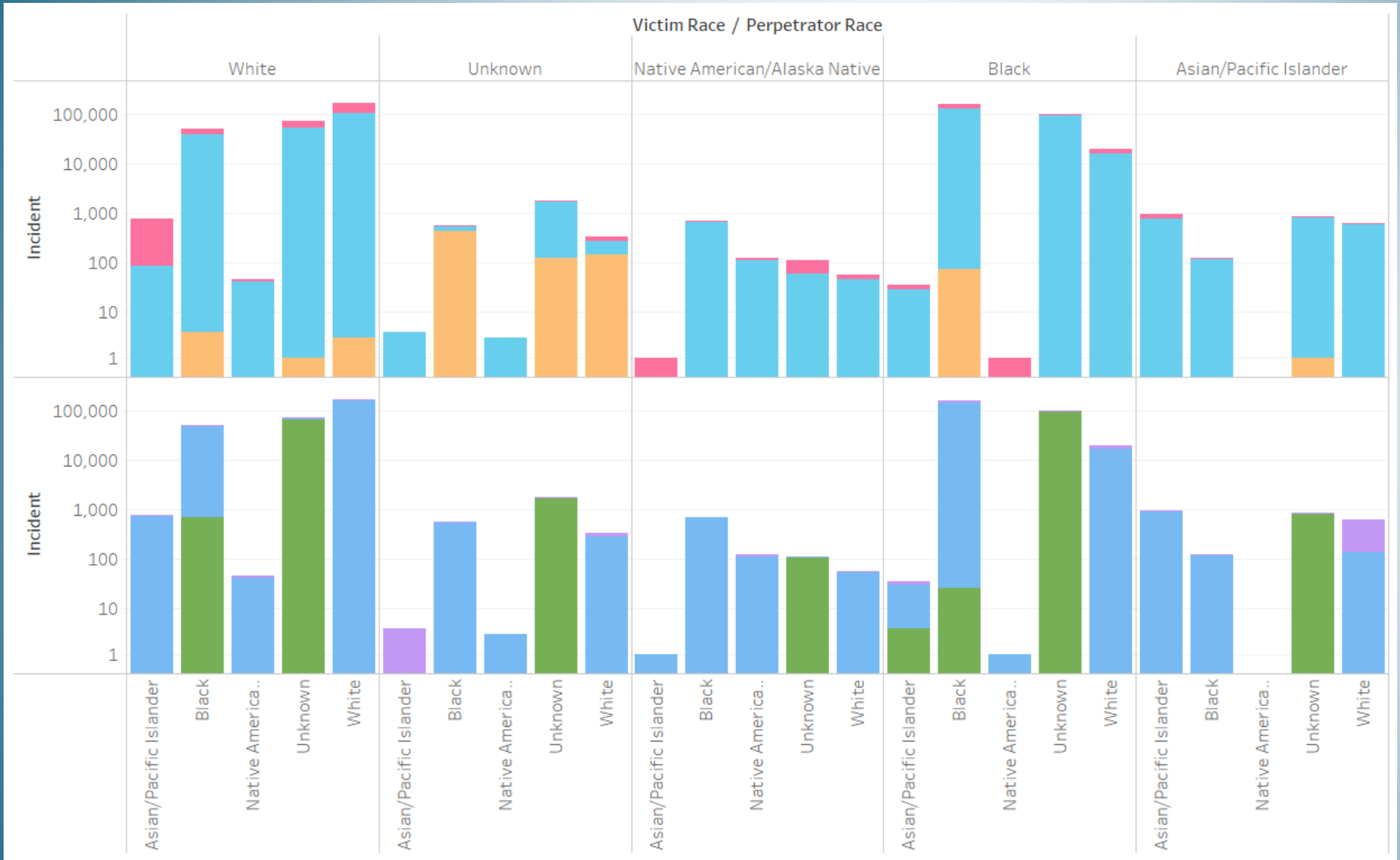
- **Peak Age Group:** Highest victimization in **20-29 years**, followed by **30-39 years**.
- **Declining Trend:** Victim count decreases with age after 40.
- **Least Affected: 0-9 and 90-100** age groups have the lowest victim count.
- **Reasons:** Young adults face higher exposure to risky environments, while crimes against the elderly may be underreported.
- **Implications:** Need for targeted crime prevention, awareness programs, and community safety initiatives.

Weapons used for top crime type:

- **Murder or Manslaughter** is the most frequent crime (99,736 cases).
- **Handguns** are the most commonly used weapon.
- **Knives, firearms, and blunt objects** are also frequently involved.
- **Poison, drowning, and suffocation** are rare causes.
- Highlights the need for **stricter weapon control** and crime prevention.

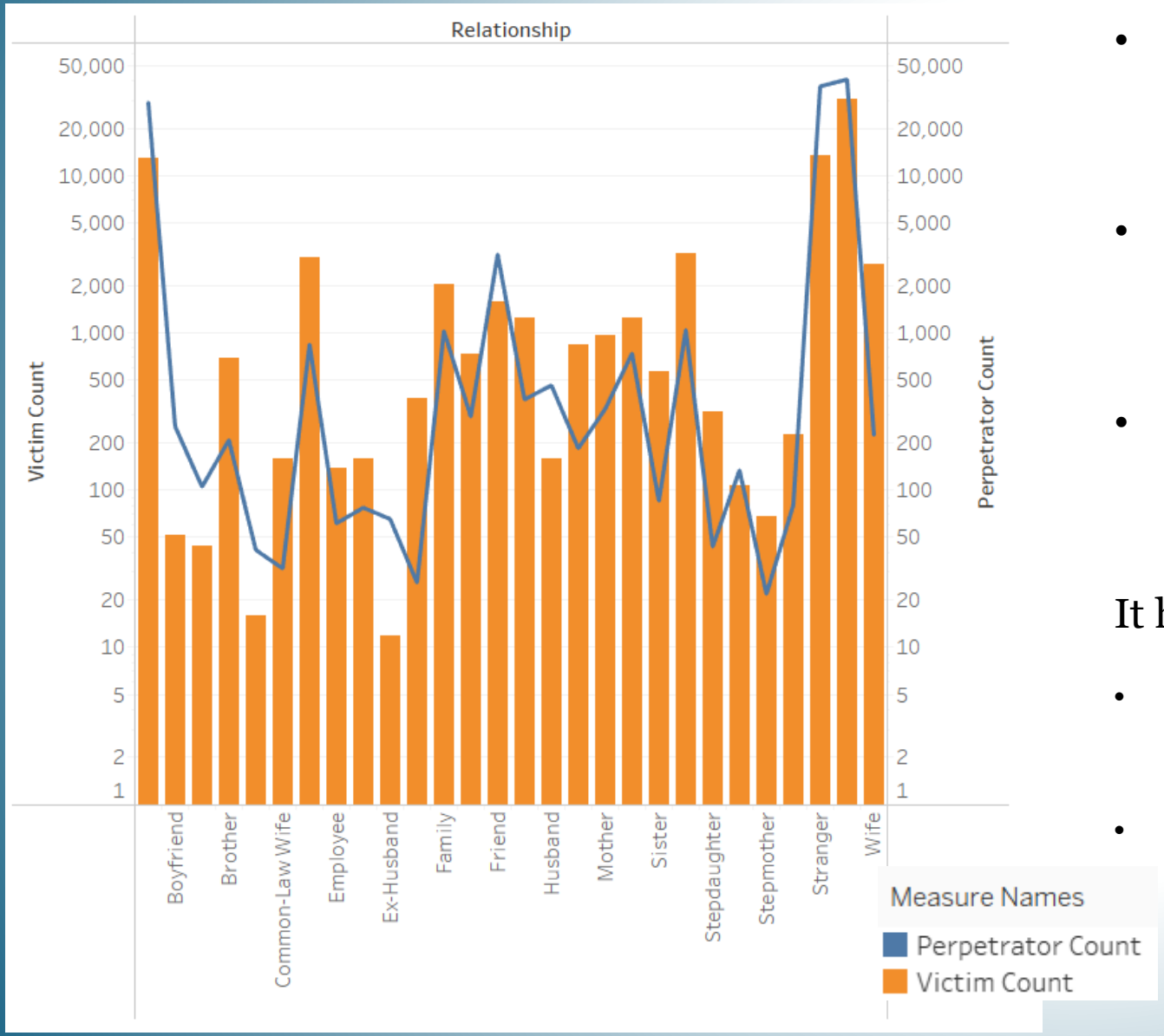


Race and perpetrator race



- **Same Race Crimes Dominate** – Most incidents occur within the same racial group.
- **White & Black Groups Lead** – Highest crime counts due to population size and reporting trends.
- **Lower Counts in Other Groups** – Fewer incidents among Asian/Pacific Islanders & Native Americans.

Relationship between Victim and perpetrator:



- **Strangers & Wives Most Affected** – High crime rates involve **strangers** and **wives**, highlighting public safety and domestic violence risks.
- **Domestic Violence is Significant** – Crimes involving **partners and family** show the need for stronger intervention.
- **Workplace & Friend-Related Crimes Exist** – Conflicts in **workplaces and friendships** also contribute to crime.

It helps in:

- **Policy making:** Strengthen **domestic violence laws** and victim support.
- **Law Enforcement:** Focus on **early intervention** and **self-defense programs**.

Time of Crime (months):

Based on the data structure, we can infer that certain crimes may have **seasonal patterns**, such as:

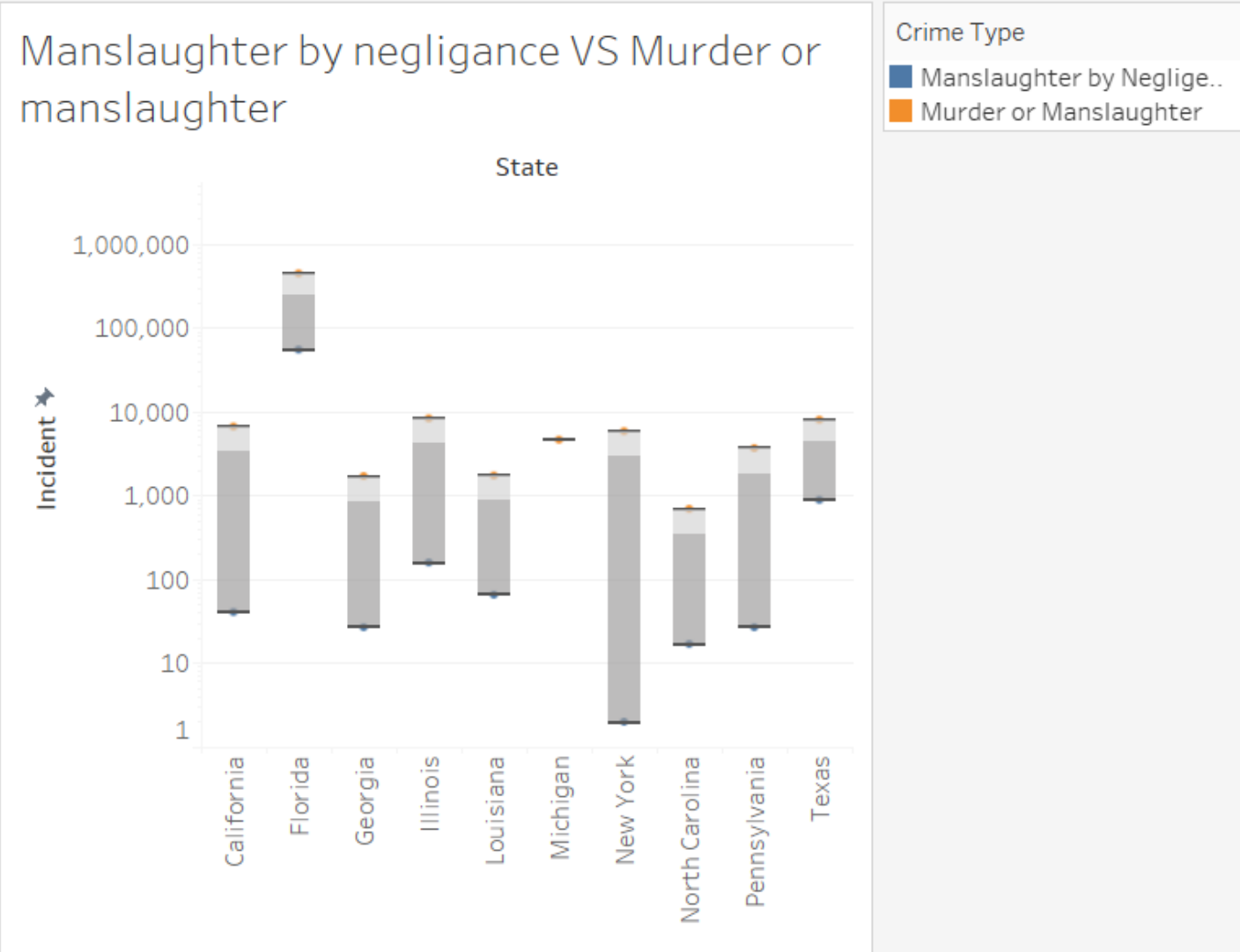
- **Summer months (June - August):** Higher crime rates due to increased outdoor activity.
- **Winter months (December - February):** Potential decrease in certain crimes but an increase in others (e.g., domestic crimes).

Insights Through Visualization:

- **Heatmaps:** Crime density visualization across U.S. states.
- **Line Charts:** Relationship between victim and prepetrator.
- **Bar Chart:** Crime trends over multiple years.
- **Interactive Dashboards:** Filtering crime data by type, year, and region.
- **Example Insights:**
- Some crimes are concentrated in urban areas.
- Certain crimes correlate with economic conditions.
- Crime spikes in specific months, possibly due to seasonal factors.

Comparative analysis

- **Urban vs. Rural Crime Rates:**
  - Comparing crime patterns in metropolitan areas vs. rural regions.
- **Socio-Economic Influence:**
  - Impact of factors like income level, education, and unemployment on crime.
- **Crime Type Comparison:**
  - Analyzing trends in violent crimes (murder, assault) vs. property crimes (theft, burglary).
- **Cross-State Comparison:**
  - Why do some states have significantly lower crime rates?
  - Impact of local policies and law enforcement practices.



To conclude:

- **Crime Rates Are Not Uniform:** Some states and cities have significantly higher crime rates than others.
- **Seasonal Trends Exist:** Certain crimes spike in summer months.
- **Economic Factors Play a Role:** Higher unemployment correlates with an increase in property crimes.
- **Violent vs. Property Crimes:** Different trends and influencing factors for each.
- **Unexpected Observations:** Any anomalies or surprising trends found in the data.

Recommendations:

- **Policy & Law Enforcement Recommendations:**
  - Allocate more law enforcement resources to high-crime areas.
  - Community-based crime prevention programs.
  - Economic interventions to reduce crime (job programs, education initiatives).

- **Future Research Suggestions:**
  - Incorporating more variables (weather, policing strategies, education levels).
  - Predictive modeling for crime forecasting.
  - Deeper regional case studies to understand anomalies.

Thank you!